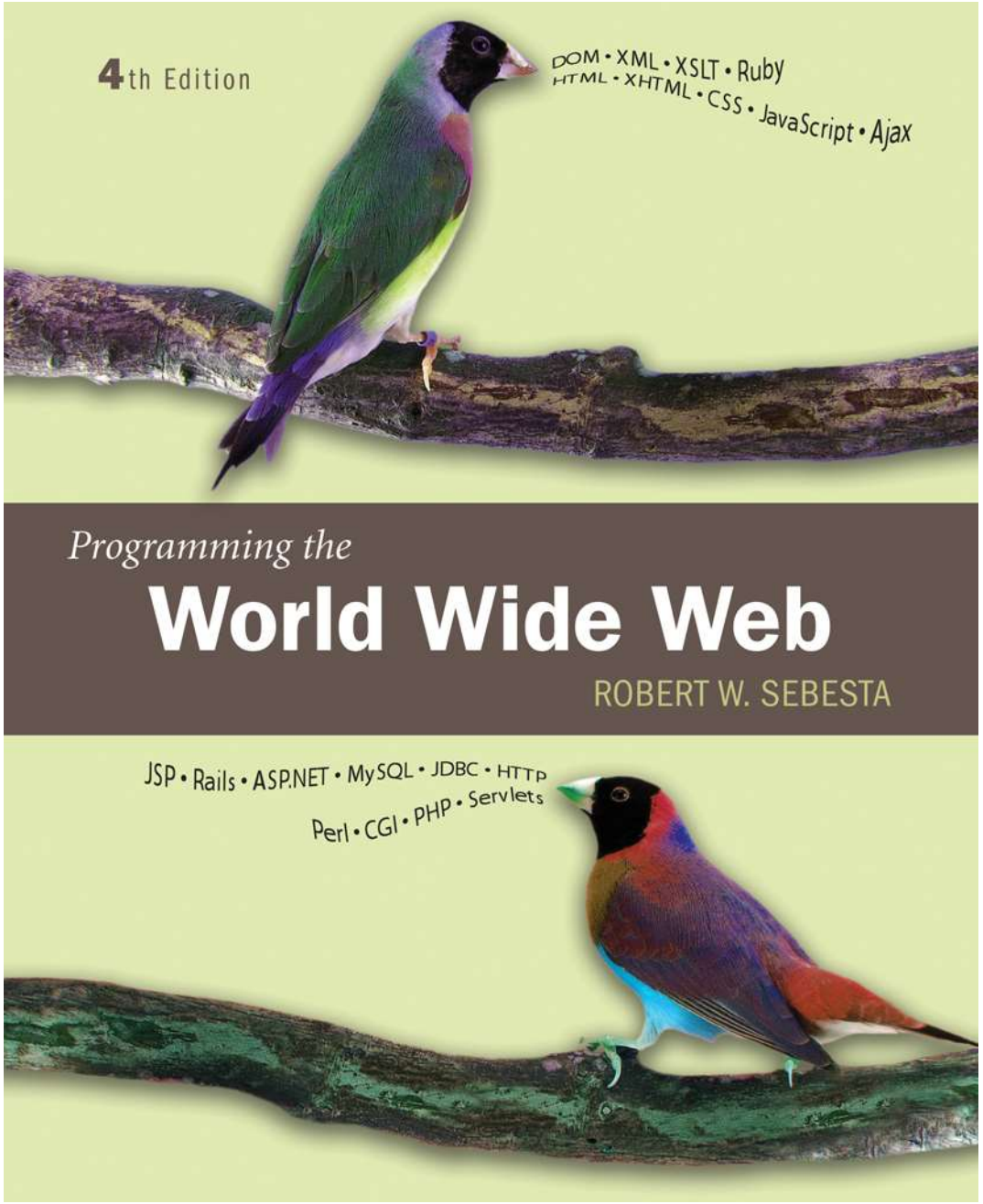


Chapter 2
Introduction to XHTML

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2.1 Origins and Evolution of HTML and XHTML

- HTML was defined with SGML (Standard Generalized Markup Language)
- Original intent of HTML: General layout of documents that could be displayed by a wide variety of browsers
- Recent versions:
  - HTML 4.0 – 1997
    - » Introduced many new features and deprecated many older features
  - HTML 4.01 - 1999 - A cleanup of 4.0
  - XHTML 1.0 - 2000
    - » Just 4.01 defined using XML, instead of SGML
  - XHTML 1.1 – 2001
    - » Modularized 1.0, and drops frames
    - » We'll stick to 1.1, except for frames

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2.1 Origins and Evolution of HTML and XHTML (continued)

- Reasons to use XHTML, rather than HTML:

  1. HTML has lax syntax rules, leading to sloppy and sometime ambiguous documents

     – XHTML syntax is much more strict, leading to clean and clear documents in a standard form

  2. HTML processors do not even enforce the few syntax rules that do exist in HTML

  3. The syntactic correctness of XHTML documents can be validated

2.2 Basic Syntax

- Elements are defined by tags (categories of content)
  - Tag format:
    - » Opening tag: <name>
    - » Closing tag: </name>
  - The opening tag and its closing tag together specify a container for the *content* they enclose

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2.2 Basic Syntax (continued)

- Not all tags have content
  - If a tag has no content, its form is <name />
- The container and its content together are called an *element*
- If a tag has attributes, they appear between its name and the right bracket of the opening tag
- Comment form: <!-- … -->
- Browsers ignore comments, unrecognizable tags, line breaks, multiple spaces, and tabs
- Tags are suggestions to the browser

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2.3 XHTML Document Structure

- Every XHTML document must begin with an xml declaration element:

```
<?xml version = "1.0"?>
<!DOCTYPE html PUBLIC "-//w3c//DTD XHTML 1.1//EN"
  http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd>
```

- `<html>`, `<head>`, `<title>`, and `<body>` are required in every document
- The whole document must have `<html>` as its root
- `html` must have the `xmlns` attribute:

```
<html xmlns = "http://www.w3.org/1999/xhtml"
```

- A document consists of a head and a body
- The `<title>` tag is used to give the document a title, which is normally displayed in the browser's window title bar (at the top of the display)
- Prior to XHTML 1.1, a document could have either a body or a frameset

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2.4 Basic Text Markup

- Text is normally placed in paragraph elements

1. Paragraph Elements

- The <p> tag breaks the current line and inserts a blank line - the new line gets the beginning of the content of the paragraph
- The browser puts as many words of the paragraph's content as will fit in each line

```
<?xml version = "1.0"?>
  <!DOCTYPE html PUBLIC "-//w3c//DTD XHTML 1.1//EN"
     http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd>
<!-- greet.hmtl
     A trivial document
     -->
<html xmlns = "http://www.w3.org/1999/xhtml">
  <head> <title> Our first document </title>
  </head>
  <body>
    <p>
      Greetings from your Webmaster!
    </p>
  </body>
</html>
```

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2.4 Basic Text Markup (continued)

2. **Line breaks**
   - The effect of the `<br />` tag is the same as that of `<p>`, except for the blank line
     - » No closing tag!

- Example of paragraphs and line breaks

```
On the plains of hesitation <p> bleach the
bones of countless millions </p> <br />
who, at the dawn of victory <br /> sat down
to wait, and waiting, died.
```

- Typical display of this text:

```
On the plains of hesitation

bleach the bones of countless millions
who, at the dawn of victory
sat down to wait, and waiting, died.
```

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2.4 Basic Text Markup (continued)

3. Preserving white spaces

Using the tag <pre> </pre> the whitespaces can be preserved

Example:

```
<p><pre>
RVCE
      CSE
                DEPT
</pre></p>
```

4. Headings

- Six sizes, 1 - 6, specified with <h1> to <h6>
- 1, 2, and 3 use font sizes that are larger than the default font size
- 4 uses the default size
- 5 and 6 use smaller font sizes

Example Headings.html

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2.4 Basic Text Markup (continued)

5. Blockquotes

- Content of <blockquote>
- To set a block of text off from the normal flow and appearance of text
- Browsers often indent, and sometimes italicize
- <blockquote> aa </ blockquote>

6. Font Styles and Sizes (can be nested)

- Boldface - <b>
- Emphasis -
- Strong -
- Superscript - <sup>
- Subscript - <sub>

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2.4 Basic Text Markup (continued)

Example: x₂³

Display: x_2^3

7. **Character Entities**

Char.	*Entity*	*Meaning*
&	&	Ampersand
<	<	Less than
>	>	Greater than
”	"	Double quote
’	'	Single quote
¼	¼	One quarter
½	½	One half
¾	¾	Three quarters
°	°	Degree
(space)		Non-breaking space

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2.4 Basic Text Markup (continued)

8. Horizontal rules

- <hr /> draws a line across the display, after a line break

9. **meta** Element

- For search engines to categorize the document
- Used to provide additional information about a document, with attributes, not content

Example:

<meta name = “Title” content = “3 mistakes of my life” />

- W3C HTML Validation Service

`http://validator.w3.org/file-upload.html`

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2.5 Images

- GIF (Graphic Interchange Format)
  - 8-bit color (256 different colors)
- JPEG (Joint Photographic Experts Group)
  - 24-bit color (16 million different colors)
- Both use compression, but JPEG compression is better
- Images are inserted into a document with the `<img />` tag with the `src` attribute
  - The `alt` attribute is required by XHTML
    - » Purposes:
      1. Non-graphical browsers
      2. Browsers with images turned off

```
<img src = "comets.jpg"
     alt = "Picture of comets" />
```

- The `<img>` tag has 30 different attributes, including `width` and `height` (in pixels)
- Portable Network Graphics (PNG)
  - Relatively new
  - Should eventually replace both gif and jpeg

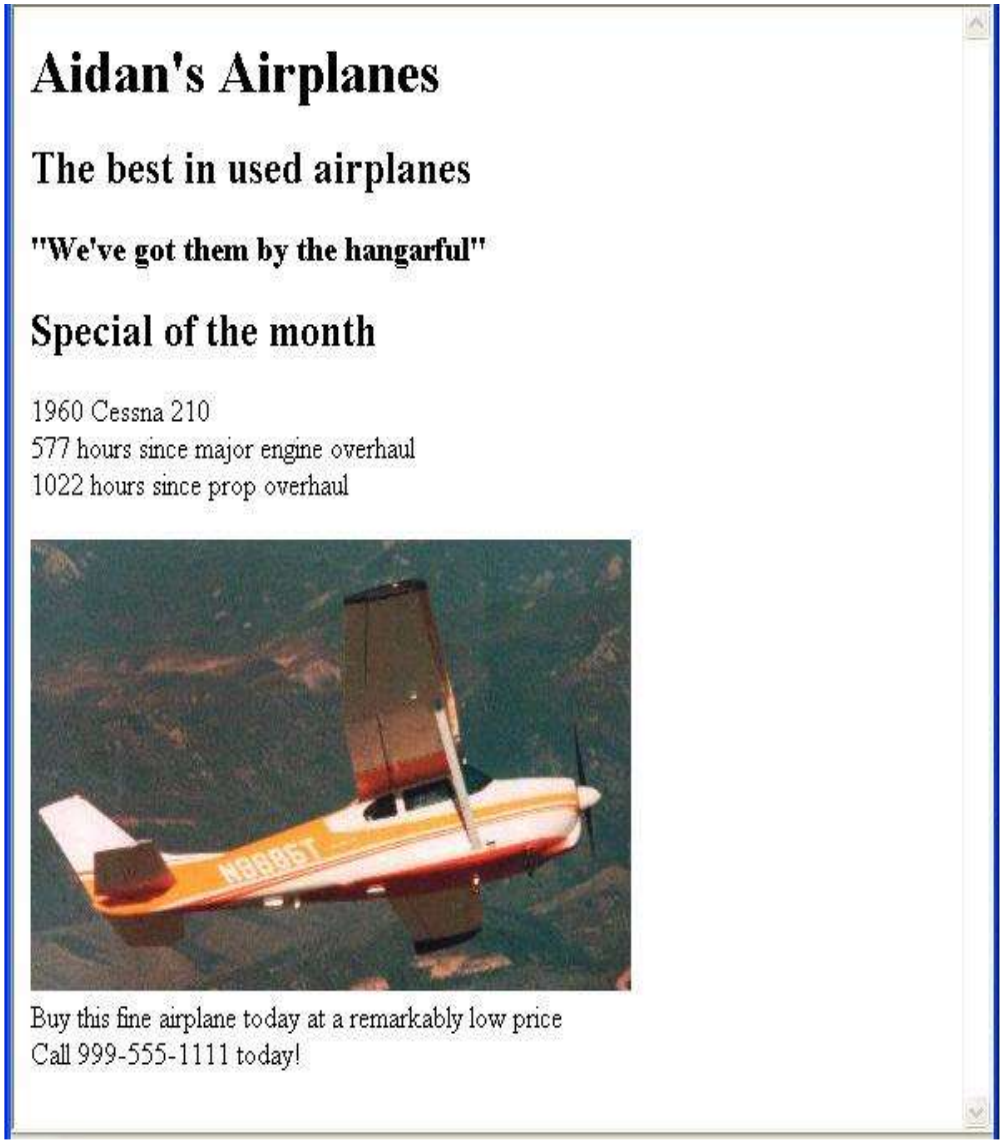
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2.5 Images (continued)

```
<!-- image.html
     An example to illustrate an image
     -->
<html xmlns = "http://www.w3.org/1999/xhtml">
  <head> <title> Images </title>
  </head>
  <body>
    <h1> Aidan's Airplanes </h1>
    <h2> The best in used airplanes </h2>
    <h3> "We've got them by the hangarful"
    </h3>
    <h2> Special of the month </h2>
    <p>
      1960 Cessna 210 <br />
      577 hours since major engine overhaul
      <br />
      1022 hours since prop overhaul
      <br /><br />
      <img src = "c210new.jpg"
           alt = "Picture of a Cessna 210"/>
      <br />
      Buy this fine airplane today at a
      remarkably low price <br />
      Call 999-555-1111 today!
    </p>
  </body>
</html>
```

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2.5 Images (continued)



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2.6 Hypertext Links

2.6.1 Links

- Hypertext is the essence of the Web!
- A link is specified with the `href` (*h*ypertext *ref*erence) attribute of `<a>` (the anchor tag)
  - The content of `<a>` is the visual link in the document
  - If the target is a whole document (not the one in which the link appears), the target need not be specified in the target document as being the target
- Note: Relative addressing of targets is easier to maintain and more portable than absolute addressing

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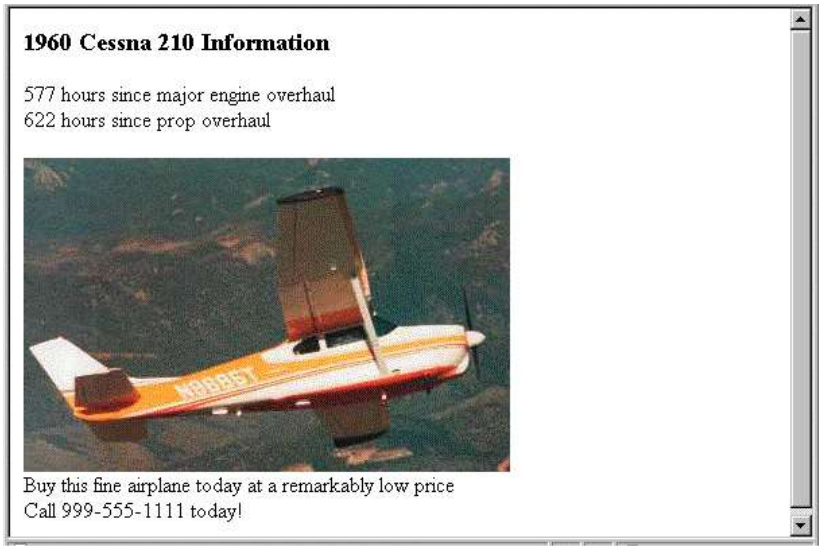
2.6 Hypertext Links (continued)

```
<!-- link.html
     An example to illustrate a link
     -->
<html xmlns = "http://www.w3.org/1999/xhtml">
  <head> <title> Links </title>
  </head>
  <body>
    <h1> Aidan's Airplanes </h1>
    <h2> The best in used airplanes </h2>
    <h3> "We've got them by the hangarful"
    </h3>
    <h2> Special of the month </h2>
    <p>
      1960 Cessna 210 <br />
      <a href = "C210data.html">
        Information on the Cessna 210 </a>
    </p>
  </body>
</html>
```

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2.6 Hypertext Links (continued)





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2.6 Hypertext Links (continued)

2.6.2 Target within documents

- If the target is not at the beginning of the document, the target spot must be marked
- Target labels can be defined in many different tags with the `id` attribute, as in

```
<h1 id = "baskets"> Baskets </h1>
```

- The link to an `id` must be preceded by a pound sign (`#`); If the `id` is in the same document, this target could be

```
<a href = "#baskets">
                    What about baskets? </a>
```

- If the target is in a different document, the document reference must be included

```
<a href = "myAd.html#baskets”> Baskets </a>
```

- Style note: a link should blend in with the surrounding text, so reading it without taking the link should not be made less pleasant
- Links can have images:

```
<a href = "c210data.html“>
  <img src = "smallplane.jpg"
    alt = "Small picture of an airplane " />
      Info on C210 </a>
```

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2.7 Lists

- *Unordered lists*
- The list is the content of the `<ul>` tag
- List elements are the content of the `<li>` tag

```
<h3> Some Common Single-Engine Aircraft </h3>
 <ul>
   <li> Cessna Skyhawk </li>
   <li> Beechcraft Bonanza </li>
   <li> Piper Cherokee </li>
 </ul>
```

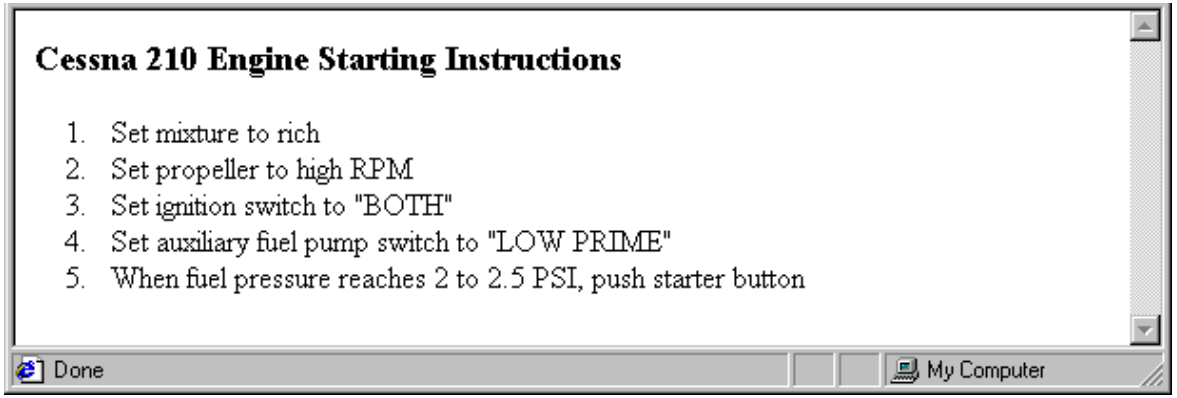


- *Ordered lists*
  - The list is the content of the `<ol>` tag
  - Each item in the display is preceded by a sequence value

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2.7 Lists (continued)

```
<h3> Cessna 210 Engine Starting Instructions
</h3>
<ol>
  <li> Set mixture to rich </li>
  <li> Set propeller to high RPM </li>
  <li> Set ignition switch to "BOTH" </li>
  <li> Set auxiliary fuel pump switch to
       "LOW PRIME" </li>
  <li> When fuel pressure reaches 2 to 2.5
       PSI, push starter button </li>
</ol>
```



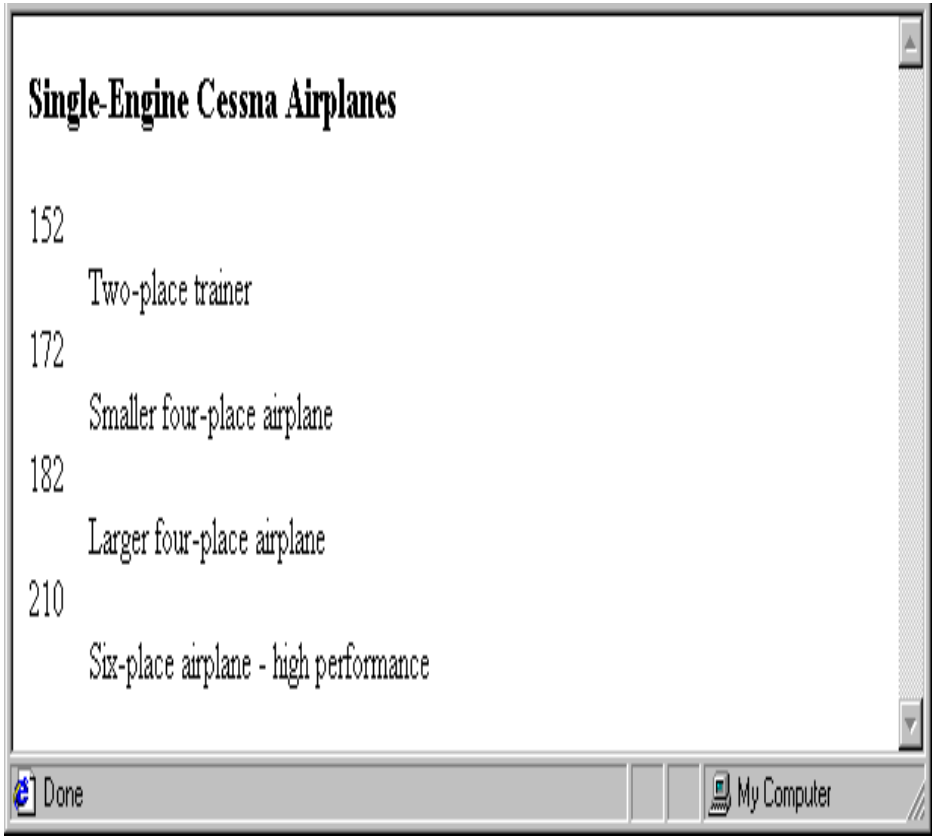
- *Nested lists*
  - Any type list can be nested inside any type list
  - The nested list must be in a list item

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2.7 Lists (continued)

- *Definition lists (for glossaries, etc.)*
  - List is the content of the `<dl>` tag
  - Terms being defined are the content of the `<dt>` tag
  - The definitions themselves are the content of the `<dd>` tag

```
<h3> Single-Engine Cessna Airplanes </h3>
<dl >
  <dt> 152 </dt>
  <dd> Two-place trainer </dd>
  <dt> 172 </dt>
  <dd> Smaller four-place airplane </dd>
  <dt> 182 </dt>
  <dd> Larger four-place airplane </dd>
  <dt> 210 </dt>
  <dd> Six-place airplane - high performance
  </dd>
</dl>
```



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2.8 Tables

- A table is a matrix of cells, each possibly having content
- The cells can include almost any element
- Some cells have row or column labels and some have data
- A table is specified as the content of a `<table>` tag
- A `border` attribute in the `<table>` tag specifies a border between the cells
- If `border` is set to `"border"`, the browser's default width border is used
- The `border` attribute can be set to a number, which will be the border width
- Without the `border` attribute, the table will have no lines!
- Tables are given titles with the `<caption>` tag, which can immediately follow `<table>`

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2.8 Tables (continued)

- Each row of a table is specified as the content of a `<tr>` tag
- The row headings are specified as the content of a `<th>` tag
- The contents of a data cell is specified as the content of a `<td>` tag

```
<table border = "border">
   <caption> Fruit Juice Drinks </caption>
     <tr>
       <th> </th>
       <th> Apple </th>
       <th> Orange </th>
       <th> Screwdriver </th>
     </tr>
     <tr>
       <th> Breakfast </th>
       <td> 0 </td>
       <td> 1 </td>
       <td> 0 </td>
     </tr>
     <tr>
       <th> Lunch </th>
       <td> 1 </td>
       <td> 0 </td>
       <td> 0 </td>
     </tr>
   </table>
```

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2.8 Tables (continued)



- A table can have two levels of column labels
  - If so, the `colspan` attribute must be set in the `<th>` tag to specify that the label must span some number of columns

```
<tr>
  <th colspan = "3"> Fruit Juice Drinks </th>
</tr>
<tr>
  <th> Orange </th>
  <th> Apple </th>
  <th> Screwdriver </th>
</tr>
```



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2.8 Tables (continued)

- If the rows have labels and there is a spanning column label, the upper left corner must be made larger, using `rowspan`

```
<table border = "5">
  <tr>
    <td rowspan = "2"> </td>
    <th colspan = "3"> Fruit Juice Drinks
    </th>
  </tr>
  <tr>
    <th> Apple </th>
    <th> Orange </th>
    <th> Screwdriver </th>
  </tr>
</table>
```

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2.8 Tables (continued)

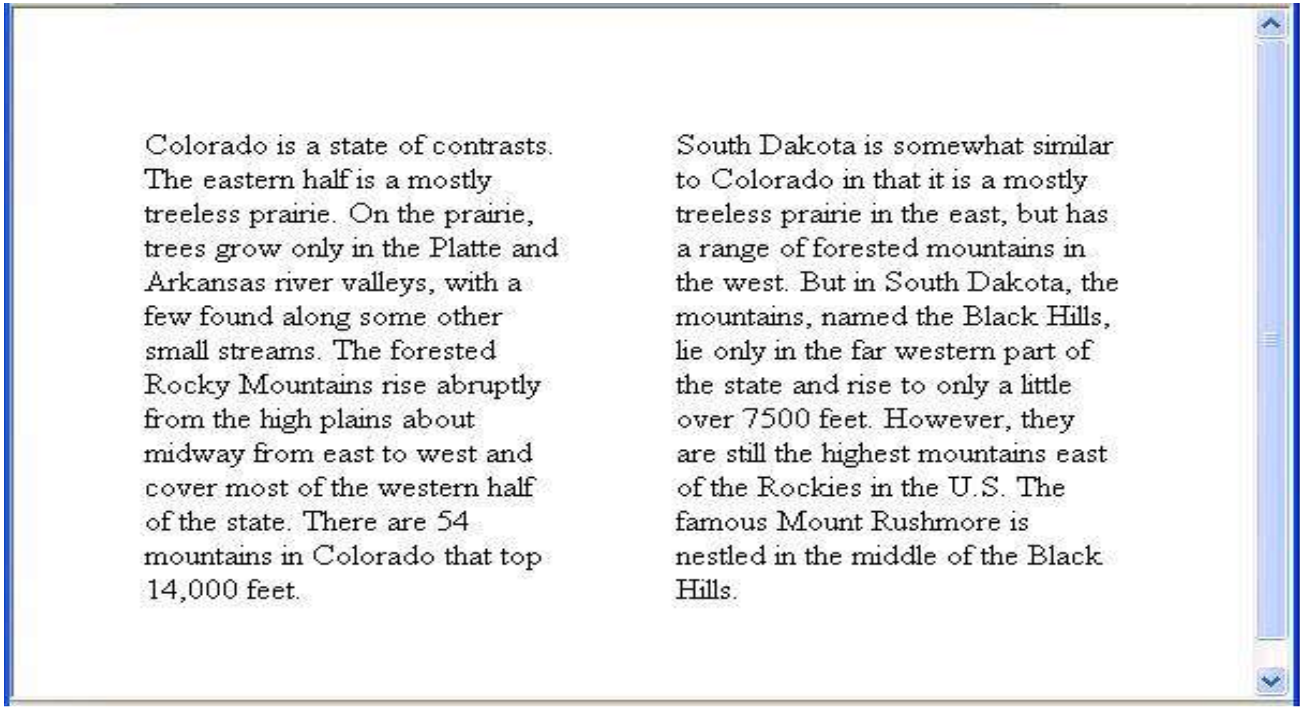
- The `align` attribute controls the horizontal placement of the contents in a table cell
  - Values are `left`, `right`, and `center` (default)
  - `align` is an attribute of `<tr>`, `<th>`, and `<td>` elements
- The `valign` attribute controls the vertical placement of the contents of a table cell
  - Values are `top`, `bottom`, and `center` (default)
  - `valign` is an attribute of `<th>` and `<td>` elements

  → SHOW `cell_align.html` and display it
- The `cellspacing` attribute of `<table>` is used to specify the distance between cells in a table
- The `cellpadding` attribute of `<table>` is used to specify the spacing between the content of a cell and the inner walls of the cell

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2.8 Tables (continued)

```
<table cellspacing = "50">
     <tr>
        <td> Colorado is a state of …
        </td>
        <td> South Dakota is somewhat…
        </td>
     </tr>
  </table>
```



- *Table Sections*
  - Header, body, and footer, which are the elements: `thead`, `tbody`, and `tfoot`
  - If a document has multiple `tbody` elements, they are separated by thicker horizontal lines

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2.9 Forms

- A form is the usual way information is gotten from a browser to a server
- XHTML has tags to create a collection of objects that implement this information gathering
  - The objects are called *widgets* (e.g., radio buttons and checkboxes)
- When the Submit button of a form is clicked, the form's values are sent from the web browser to the web server
- All of the widgets, or components of a form are defined in the content of a `<form>` tag
  - The only required attribute of `<form>` is `action`, which specifies the URL of the application that is to be called when the Submit button is clicked

```
action =
  "http://www.cs.ucp.edu/cgi-bin/survey.pl"
```

    - » If the form has no action, the value of `action` is the empty string

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2.9 Forms (continued)

- The `method` attribute of `<form>` specifies one of the two possible techniques of transferring the form data to the server, `get` or `post`
  - `get` is the default
- In both cases the form data is coded into a text string when the user clicks the submit button
- *Widgets*
  - Many are created with the `<input>` tag
    - » The `type` attribute of `<input>` specifies the kind of widget being created

    1. Text
    - Creates a horizontal box for text input
    - Default size is 20; it can be changed with the `size` attribute
    - If more characters are entered than will fit, the box is scrolled (shifted) left

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2.9 Forms (continued)

```
Example: If you enter
                This is web programming class
  The text box collects whole string but the string would be
   scrolled to the right leaving the following shown in the
   box
                eb programming class
```

- If you don't want to allow the user to type more characters than will fit, set `maxlength`, which causes excess input to be ignored

```
<input type = "text" name = "name"
       size = "25" maxlength = "25" />
```

2. Checkboxes - to collect multiple choice input

- Every checkbox requires a `value` attribute, which is the widget's value in the form data when the checkbox is 'checked'
  - » A checkbox that is not 'checked' contributes no value to the form data
- By default, no checkbox is initially 'checked'
- To initialize a checkbox to 'checked', the `checked` attribute must be set to `"checked"`

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2.9 Forms (continued)

- *Widgets* (continued)

```
Grocery Checklist
<form action = "">
  <p>
  <input type = "checkbox"  name ="groceries"
         value = "milk"  checked = "checked">
   Milk
  <input type = "checkbox"  name ="groceries"
         value = "bread">
   Bread
  <input type = "checkbox"  name = "groceries"
         value= "eggs">
   Eggs
  </p>
</form>
```



3. *Radio Buttons* - collections of checkboxes in which only one button can be 'checked' at a time
   - » Every button in a radio button group MUST have the same name

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2.9 Forms (continued)

- *Widgets* (continued)

3. *Radio Buttons* (continued)
   - If no button in a radio button group is ‘pressed’, the browser often ‘presses’ the first one

```
Age Category
<form action = "">
  <p>
  <input type = "radio"  name = "age"
   value = "under20" checked = "checked"> 0-19
  <input type = "radio"  name = "age"
         value = "20-35"> 20-35
  <input type = "radio"  name = "age"
         value = "36-50"> 36-50
  <input type = "radio"  name = "age"
         value = "over50"> Over 50
  </p>
</form>
```

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2.9 Forms (continued)



4. Menus - created with `<select>` tags

- There are two kinds of menus, those that behave like checkboxes and those that behave like radio buttons (the default)
  - Menus that behave like checkboxes are specified by including the `multiple` attribute, which must be set to `"multiple"`
- The `name` attribute of `<select>` is required
- The `size` attribute of `<select>` can be included to specify the number of menu items to be displayed (the default is 1)
  - If `size` is set to > 1 or if `multiple` is specified, the menu is displayed as a pop-up menu

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2.9 Forms (continued)

- *Widgets* (continued)

4. Menus (continued)

- Each item of a menu is specified with an `<option>` tag, whose pure text content (no tags) is the value of the item
- An `<option>` tag can include the `selected` attribute, which when assigned `"selected"` specifies that the item is preselected

```
Grocery Menu - milk, bread, eggs, cheese
<form action = "">
  <p>
    With size = 1 (the default)
    <select name = "groceries">
      <option> milk </option>
      <option> bread </option>
      <option> eggs </option>
      <option> cheese </option>
    </select>
  </p>
</form>
```

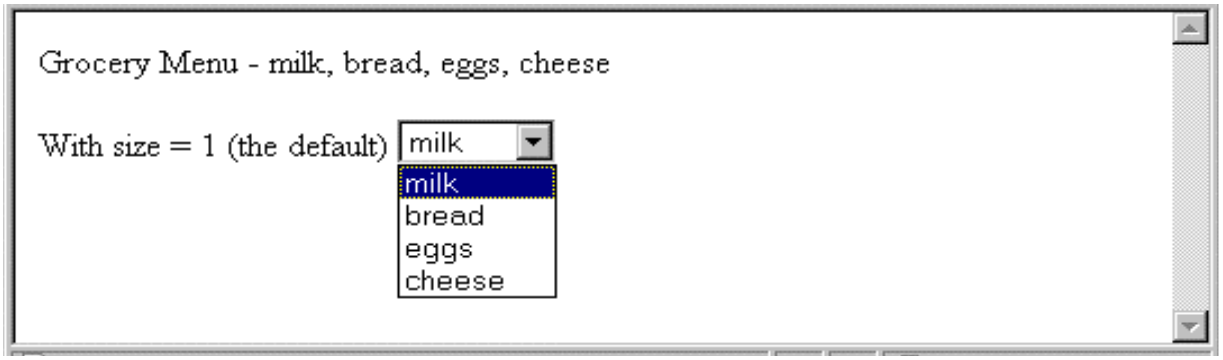
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2.9 Forms (continued)

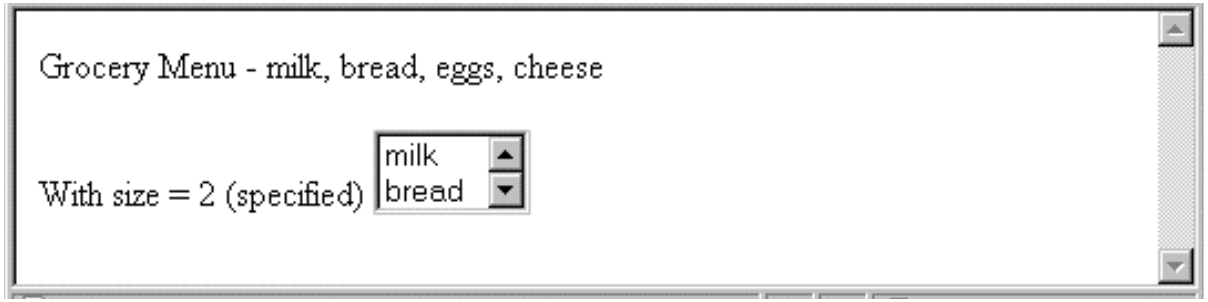
- *Widgets* (continued)



- After clicking the menu:



- After changing `size` to 2:



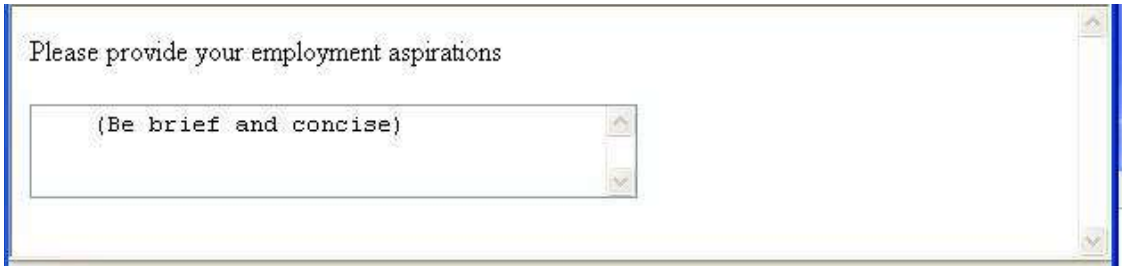
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2.9 Forms (continued)

- *Widgets* (continued)

5. Text areas - created with `<textarea>`
   - Usually include the `rows` and `cols` attributes to specify the size of the text area
   - Default text can be included as the content of `<textarea>`
   - Scrolling is implicit if the area is overfilled

```
Please provide your employment aspirations
<form action = "">
    <p>
    <textarea name = "aspirations"  rows = "3"
              cols = "40">
    (Be brief and concise)
    </textarea>
    </p>
</form>
```



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2.9 Forms (continued)

- *Widgets* (continued)

6. Reset and Submit buttons

  - Both are created with `<input>`

```
<input type = "reset"  value = "Reset Form">

<input type = "submit"  value = "Submit Form">
```

- Submit has two actions:
  1. Encode the data of the form
  2. Request that the server execute the server-resident program specified as the value of the `action` attribute of `<form>`
  - A Submit button is required in every form

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2.10 Frames

- Frames are rectangular sections of the display window, each of which can display a different document
- Because frames are no longer part of XHTML, you cannot validate a document that includes frames
- The `<frameset>` tag specifies the number of frames and their layout in the window
  - `<frameset>` takes the place of `<body>`
  - Cannot have both!
  - `<frameset>` must have either a `rows` attribute or a `cols` attribute, or both (usually the case)
  - Default is 1
  - The possible values for `rows` and `cols` are numbers, percentages, and asterisks
    - A number value specifies the row height in pixels - Not terribly useful!
    - A percentage specifies the percentage of total window height for the row - Very useful!

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2.10 Frames (continued)

- An asterisk after some other specification gives the remainder of the height of the window
- Examples:

```
<frameset rows = "150, 200, 300">

<frameset rows = "25%, 50%, 25%">

<frameset rows = "50%, 20%, *" >

<frameset rows = "50%, 25%, 25%"
          cols = "40%, *">
```

- The `<frame>` tag specifies the content of a frame
- The first `<frame>` tag in a `<frameset>` specifies the content of the first frame, etc.
  - Row-major order is used
  - Frame content is specified with the `src` attribute
  - Without a `src` attribute, the frame will be empty (such a frame CANNOT be filled later)
- If `<frameset>` has fewer `<frame>` tags than frames, the extra frames are empty

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2.10 Frames (continued)

- Scrollbars are implicitly included if needed (they are needed if the specified document will not fit)
- If a `name` attribute is included, the content of the frame can be changed later (by selection of a link in some other frame)
- Note: the `Frameset` standard must be specified in the `DOCTYPE` declaration

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2.10 Frames (continued)

```
<!-- contents.html
     The contents of the first frame of
     frames.html, which is the table of
     contents for the second frame
     -->
<html xmlns = "http://www.w3.org/1999/xhtml">
  <head> <title> Table of Contents Frame
    </title>
  </head>
  <body>
    <h4> Fruits </h4>
    <ul>
      <li> <a href = "apples.html"
              target = "descriptions">
              apples </a>
      <li> <a href = "bananas.html"
              target = "descriptions">
              bananas </a>
      <li> <a href = "oranges.html"
              target = "descriptions">
              oranges </a>
    </ul>
  </body>
</html>
```

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2.10 Frames (continued)

- Nested frames - to divide the screen in more interesting ways

→ SHOW `nested_frames.html`

2.11 Syntactic Differences between HTML & XHTML

- Case sensitivity `eg: form tag`
- Closing tags `eg:<p>`
- Quoted attribute values `eg:“3”`
- Explicit attribute values `eg:<table border>`
- `id` and `name` attributes
- Element nesting

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